GEOPHYSICAL RESEARCH LETTERS, VOL. 12, NO. 3, PAGES 133-136, MARCH 1985

MAGNETOSTRATIGRAPHY OF THE RED SANDSTONE CREEK SECTION - VAIL, COLORADO

John D. Miller[1] and Neil D. Opdyke

Department of Geology, University of Florida

Abstract. Samples for paleomagnetic study were collected from the area of Red Sandstone Creek (Lat. 39.73°N, Long. 253.6) near Vail, Colorado, in hopes of locating the stratigraphic base of the Kiaman Reversed Interval to within a few meters. Previous workers had sampled the same upper Pennsylvanian and Permian Minturn and Maroon Formations in this section and had placed the base of the Kiaman at 91m to 213m above the Minturn Formation. The interval from 276m below to 420m above the Maroon/Minturn contact was sampled in the present study at one meter intervals where possible. Of the 171 samples collected, only 2 had near normal components. The 2 were not consecutive and the normal components were removed easily by thermal cleaning or were shown to be a viscous overprint of the present field. The previous workers were apparently misled by a lack of magnetic cleaning of their samples. The conclusion of this study is that the base of the Kiaman must be stratigraphically lower in this section. The pole position from the sampled section is Lat. 42.6°N, Long. 115.4°E, A95=2.75 for N=145 samples.

Introduction

One of the goals of paleomagnetism is the extension of the magnetostratigraphic time-scale into the Paleozoic. The extension of magnetostratigraphy into the Paleozoic is hampered by a lack of Paleozoic sea floor for analysis of magnetic anomalies and by a lack of sufficient paleomagnetic studies of Paleozoic age rocks. Of paramount importance to magnetostratigraphy in the Paleozoic is the establishment of easily identifiable zones on which to build a magnetostratigraphic time-scale. The Kiaman superchron is a long (70Ma) zone of dominantly reversed magnetization in the upper Paleozoic. The length and homogeneity of this Pennsylvanian and Permian reversed interval establish it as a Paleozoic bench mark not only for world-wide correlation but also for possible extension of the magnetostratigraphic time-scale into upper Paleozoic rocks.

Several authors (Irving and Pullaiah, 1976; Roy and Morris, 1983; Turner and Tarling, 1975) have placed the base of the Kiaman at or near the Mississippian/Pennsylvanian boundary. The study of McMahon and Strangway (1968) found a possible base to the Kiaman in the Maroon Formation in Colorado. The age of the basal normal zones which McMahon and Strangway found is upper Pennsylvanian. The study presented here was meant to resolve this conflict between these two possible ages for the base of the Kiaman.

Geologic Setting

The location for this study was the section along Red Sandstone Creek near Vail, Colorado. This was the same section in which McMahon and Strangway (1968) thought they found the base of the Kiaman represented by four normal zones. For a geologic map of the area see Tweto and Lovering, 1977.

The average strike and dip for the section was S65°W and 27°W. The Red Sandstone Creek Section is located at the nose of the Red and White Syncline. (Tweto and Lovering, 1977). The plunge of this section is assumed to be negligible, but the possibility of an error, due to the plunge, does exist.

Two formations are exposed along Red Sandstone Creek, the Minturn and the Maroon (Figure 1). The Minturn Formation (Tweto, 1958) is a coarse clastic unit that consists of intertonguing marine and nonmarine deposits. The top of the formation is the Jacque Mountain Limestone (JMLS) member which has sparse fossil assemblage (Schizodus, Permophorous, Dolorothoceras, and Domotoceras species) which is long ranging and could place it anywhere from late Desmoinesian to early Permian. Stratigraphic conditions suggest that a late Desmoinesian age is to be favored, but there is no certain control on the age of this member (Brill, 1942; Tweto, 1958; Tweto and Lovering, 1977). All of the Minturn in this location is red. The JMLS forms a prominent ridge along the west side of Red Sandstone Creek (Poelchau, 1963) and marks the boundary between the Minturn and the overlying Maroon Formations. The Maroon Formation consists of thick beds of conglomerate alternating with micaceous mudstone, siltstone, and sandstone. A few thin limestone beds are present in the formation. As the name implies, it is red to maroon in color. The Maroon Formation extends well into the Permian in age.

Sample Treatment

All of the samples were drilled in the field with a hand-held, gasoline-powered drill, and were oriented in place with a Brunton compass and an inclinometer. The samples were cut in 2.5cm segments and were stored in an unshielded room. Thermal demagnetization was accomplished by progressive heating in a Schonstedt TSD-1 Thermal Demagnetizer with a heating chamber DC field of 100 gammas and a cooling chamber DC field of less than 5 gammas. Alternating Field (AF) demagnetization was done using a

[1]Now at Lamont Doherty Geological Observ.

Copyright 1985 by the American Geophysical Union.

Paper number 4L6322.
0094-8276/85/004L-6322$03.00

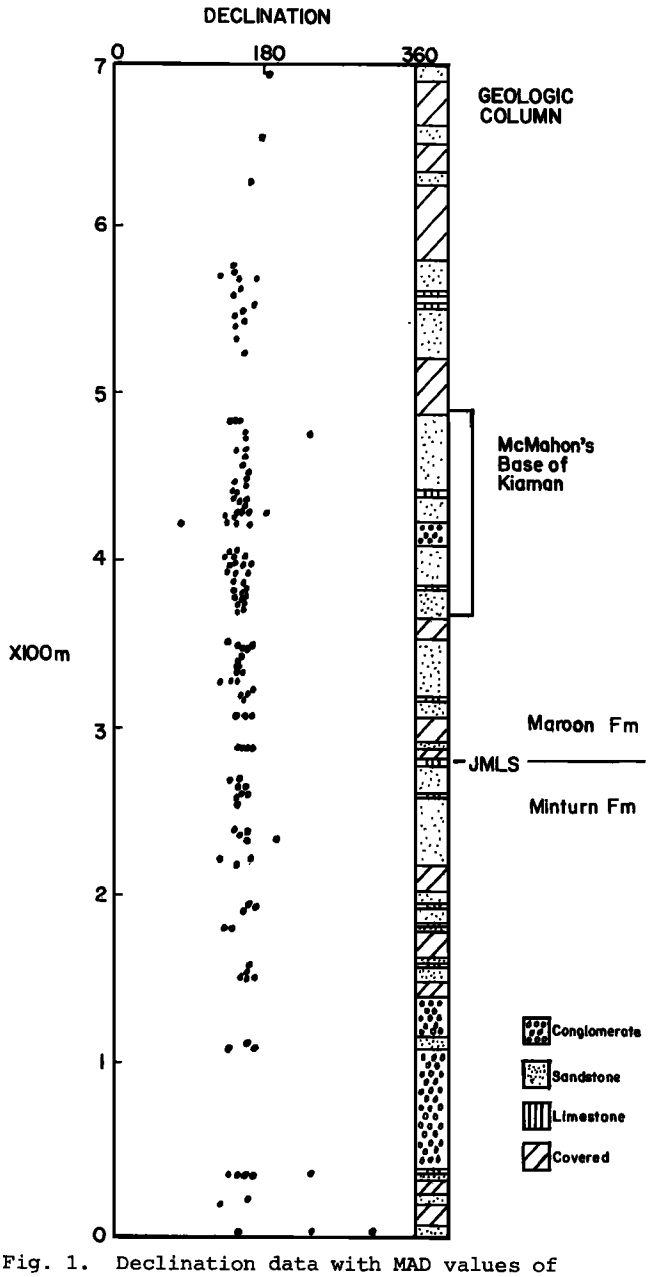


Fig. 1. Declination data with MAD values of 10° or less (tilt corrected) versus stratigraphic position and gross geologic section. Arbitrary datum.

Schondstedt GSD-1 AC Demagnetizer. Measurements were made using both a Schondstedt DSM-2 Digital Spinner Magnetometer and also a SCT Cryogenic Magnetometer.

Thermal Demagnetization

A total of 36 samples were completely demagnetized to find the thermal behaviour of the rocks. The majority of Zijderveld diagrams show consistent decay (Figure 2). A few samples showed two component behavior. The samples universally became extremely susceptable to viscous magnetization at high temperatures. This viscosity normally appeared at 650°C but some samples became viscous as low as 600°C or as high as 680°C. The viscous magnetization manifested itself as rapid decay of magnetic directions of samples within the magnetometer. The fact that these samples became viscous did not allow for a confident determination of many high-temperature components and high-temperature points of viscous samples were excluded from best fit calculations. The best fit calculations were done as described by Kirschvink, (1980). A few samples (3% of the total) never decayed to the origin. One sample (56A) showed a marked shift from an initial direction below 300 C°of declination 300°, inclination 51.8° to a direction of 235°, -8.5° above 300°C. There is no good explanation for this behavior, considering that the magnetization of a sample drilled at exactly the same level stratigraphically had a natural remanent magnetization (NRM) with a reversed direction and maintained that direction through heating to 550°C.

Alternating Field Demagnetization

Alternating Field (AF) demagnetization proved effective to remove some low coercivity components in the red beds, but not to significantly decrease the magnetic intensity or to change the magnetic direction. AF demagnetization was effective with the limestones in removing a much greater proportion of the magnetization. Most of the limestones quickly decayed to the origin with no consistency in direction.

A viscous remanent remagnetization test

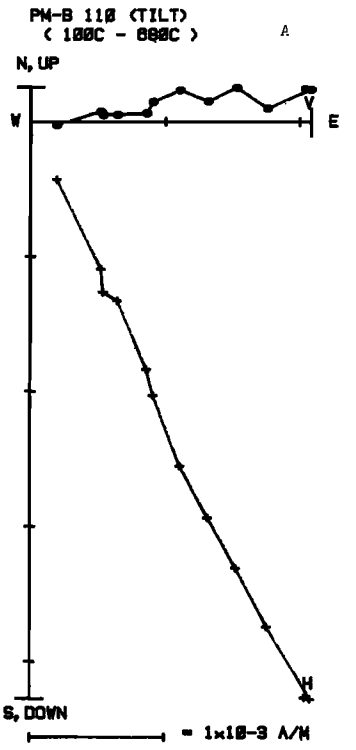


Fig. 2. Zijderveld diagram of thermal demagnetization. Points shown are [°C] 100, 200, 300, 400, 450, 500, 550, 600, 625, 650, 660, 670, 680.

19448007, 1985, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL012i003p00133 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

showed all of the limestones except for two to be overprinted entirely by the present magnetic field. The two limestones that were not viscously remagnetized had directions consistent with the red bed directions.

Cleaned Directions

Figure 3 is the stereographic plots of the magnetic directions obtained by best fit analyses of the thermal decay diagrams of samples with Maximum Angular Deviation (MAD) (Kirschvink, 1980) values of 10 or less with all viscous samples removed. Figure 1 is the same declination data plotted stratigraphically. The plots are extremely consistent and show only one point that represents near normal polarity. That one point is the low-temperature component of 56A which was previously discussed. The high inclination of this component disqualifies it from being a normal direction during the Kiaman.

The area directly around Red Sandstone Creek was not suitable for a fold test, so none was done in this study. Two previous studies (McMahon and Strangway, 1968; Christensen, 1974) have positive fold tests on the Maroon Formation that show that the magnetization predates the Laramide folding. The paleopole position of this study is also similar to other late Carboniferous and early Permian pole positions. This suggests that the magnetization of these red beds is also late Carboniferous or early Permian although later magnetization is possible.

Comparison with McMahon's Study

McMahon, in her thesis (1966), reported 4 zones of normal polarity in this section and placed the base of the Kiaman at 91m to 213m above the Jacque Mountain Limestone. The declinations from all sections of Figure 1 show remarkable consistency. The inclination data were also consistent from the top to the bottom of this section (shallow, mixed positive and negative). Although there are some gaps in the sampling, it is clear that no major normal zones could have been missed, especially in the basal zone suggested by McMahon. In McMahon's thesis (1966), there is no indication that any of the samples from Red sandstone Creek were processed with any sort of magnetic cleaning other than AF. This fact is taken to be the reason that there were no normal zones found in this study, while 4 were reported in her's. However, the pole position for this study and for McMahon's are very similar which points out that most of the samples retained directions essentially similar to their NRM's through thermal cleaning.

Pole Position

This study was not designed for the determination of a pole position, and so the pole position reported here should be regarded as tentative pending further work in the area. Fisher (1953) statistics done on the data yield: Dec. 150.6, Inc. -1.1, alpha 95 = 2.75, K = 19.2, R = 137.5, N = 145. The pole position

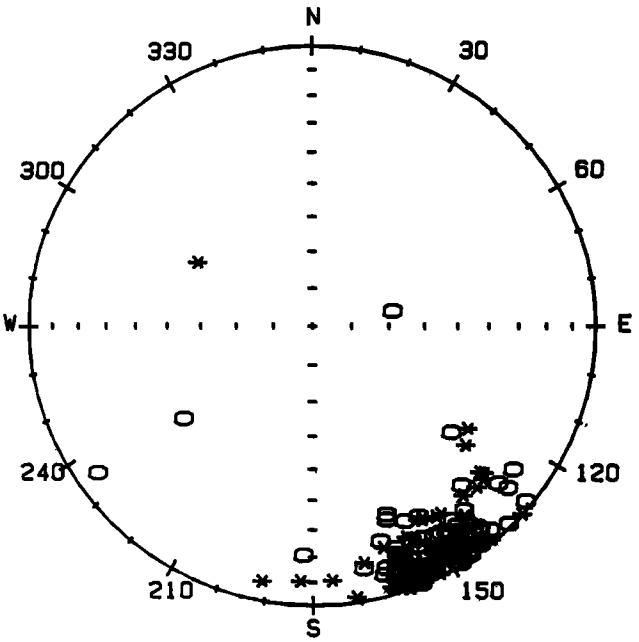


Fig. 3. Stereonet plots of thermal demagnetization directions with MAD values of 10° or less after tectonic correction. The strike and dip of the section vary by less than 10°. Insitu mean direction is dec, inc; 149.2, -29.6. Open points are positive inclination.

calculated from all of the samples in this study is: Long. 115.4°E, Lat. 42.6°N. This pole position falls near the 270Ma point on the 30Ma window of the North American Polar Wander path of Irving and Irving (1982), but lies southwest of the path proper. This fact suggests that there is some unknown structural correction required of this site. Further studies are in progress which should resolve the question of additional correction.

Conclusion

The main conclusion to be drawn from this study is that neither the base of the Kiaman nor any major normal events within the Kiaman are found in this section. This means that the base of the Kiaman is still to be found in western North America and must lie below the Red Sandstone Creek section. This study removes the inconsistency between the Kiaman base proposed by McMahon and Strangway, and that of Irving and Pullaiah (also Roy and Morris and Turner and Tarling) and supports a Desmoinesian or older age for the Kiaman base.

Acknowledgements. The authors wish to thank Dr. James Channell for help in the laboratory and in interpretation of the data.

This study was supported by a grant from the Division of Sponsored Research of the University of Florida.

References

Brill, K. G., Jr., 1942. Late Paleozoic stratigraphy of Gore area, Colorado, Am.

19448007, 1985, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL012i003p00133 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Asso. Petr. Geol. Bull., 26, 1375-97, 1942.

Christensen, F. D., A paleomagnetic investigation of the Permo-Carboniferous Maroon and upper Permian-lower Triassic State Bridge Formation in north-central Colorado. M. S. Thesis, University of Texas, Dallas, 1974.

Fisher, R. A., Dispersion on a sphere. Proc. R. Soc. Lond., A217, 295-305, 1953.

Irving, E., Irving, G. A., Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana. Geophysical Surveys, 5, 141-188, 1982.

Irving, E., and Pullaiah, G., Reversals of the geomagnetic field, magnetostratigraphy and relative magnetude of paleosecular variation in the Phanerozoic, Earth-Sci. Rev., 12, 35-64, 1976.

Kirschvink, J. L., The least-square line and plane analyses of paleomagnetic data. Geophys J. R. Astro. Soc., 62, 699-718, 1980.

McMahon, B. E., Paleomagnetic investigation of late Paleozoic and early Triassic red beds of central Colorado. University of Colorado, Ph.D. Thesis, 1966.

McMahon, B. E., and Strangway, D. W., Investigation of Kiaman magnetic division in Colorado red beds. Geophys. J. R. Astro. Soc., 15, 265-285, 1968.

Poelchau, H.S., Geology of the Gore Creek area, Eagle County, Colorado. M.S. Thesis, 76 pp., University of Colorado, 1963.

Roy, J. L., and Morris, W. A., A review of paleomagnetic results from the Carboniferous of North America; the concept of Carboniferous geomagnetic field horizon markers, Earth and Planetary Sci. Let., 65, 167-181, 1983.

Turner, P., and Tarling, D. H., Implications of new paleomagnetic results from the Carboniferous system of Britain. J. Geol. Soc. Lond., 131, 469-488, 1975.

Tweto, O., Pennsylvanian stratigraphic section in the Minturn-Pando area, Colorado, in: Symposium on Pennsylvanian Rocks of Colorado and Adjacent Areas; Rocky Mountain Association Geologists, pp. 126-128, 1958.

Tweto, O., and Lovering, T. S., Geology of the Minturn 15-minute quadrangle, Eagle & Summit Counties, Colorado. U.S.G.S. Prof. Paper 956, 96p. 1977.

J. D. Miller, Lamont-Doherty Geological Observatory, Department of Geological Sciences, Columbia University, Palisades, NY 10964.

N. D. Opdyke, University of Florida, Department of Geology, 1112 GPA Bldg., Gainesville, FL 32611.

(Received August 16, 1984
revised October 10, 1984;
accepted October 10, 1984.)

19448007, 1985, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL012i003p00133 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License